Saliency-Based SAR Target Detection via Convolutional Sparse Feature Enhancement and Bayesian Inference

Jia-Cheng Ni, Ying Luo, *Member, IEEE*, Dan Wang, Jia Liang, and Qun Zhang, *Senior Member, IEEE*

***Abstract*— Traditional synthetic aperture radar (SAR) target detection methods use matched filtered SAR images as input, and the detection performance is restricted due to the high sidelobes and speckle noise of these images. Sparse SAR imaging methods developed in recent years provide the advantages of reducing sidelobes, noise, and clutter. The imaging results obtained with these methods could help improve the SAR target detection performance. In this article, to improve the target detection performance using sparse SAR images as input, we proposed a convolutional sparse feature enhancement method to meet the needs of Bayesian saliency detection. The proposed Bayesian saliency joint target detection method comprised the following three steps: first, to obtain sparse SAR images with continuous contours and fewer holes in the target area, we proposed a convolutional L_1 sparse regularization method. Second, a regularization parameter optimization method was derived to quickly obtain optimal regularization parameters for saliency detection. Finally, target detection results were obtained through a superpixel-based Bayesian saliency joint detector. Extensive experiments verified that the proposed method could improve the SAR target detection accuracy in complex backgrounds.**

***Index Terms*— Bayesian inference, saliency detection, sparse synthetic aperture radar (SAR) feature enhancement, SAR target detection, sparse SAR images.**

I. Introduction

SYNTHETIC aperture radar (SAR) technology has been widely used in many military and civilian fields [1]. As one of the key fields in SAR applications, SAR image target detection has received considerable attention [2]. Different kinds of SAR target detection methods have been proposed over the years, including constant false alarm rate (CFAR)-based methods [3], [4], [5], saliency-based methods [6], [7], [8], [9], [10], [11], and deep learning-based methods [12], [13], [14], [15]. Among the above methods, saliency detection is an effective way to obtain high detection rates and low false alarm rates. Saliency is an essential visual description of optical images and has been studied in depth for optical image detection purposes [16], [17], [18], [19]. However, SAR images are unlike optical images in that they include significant sidelobes of targets and substantial speckle noise in the background, making typical saliency detection algorithms ineffective for SAR images. To overcome these difficulties, several saliency detection methods suitable for SAR images have been developed. Liu and Cao [6] used saliency based on multiscale singular value decomposition to suppress speckle noise in complex backgrounds. Wang et al. [7] proposed a pattern recurrence model that employed the patch-level intensity contrast instead of the pixel-level intensity contrast to overcome speckle noise. Ni et al. [8] introduced background context awareness within a multiscale framework to increase the distinction between foreground and background in SAR images. Tu and Su [9] developed a multiscale saliency detection method in the spectrum domain to segment SAR images in complex backgrounds. Wang et al. [10] proposed a Bayesian saliency detection model for SAR images. This model combined the prior map obtained via a superpixel segmentation method with two likelihood maps obtained by convex hulls, achieving satisfactory target detection results for SAR images. Recently, saliency detection has also been used in deep learning-based SAR target detection methods. Du et al. [11] proposed a saliency-guided convolutional neural network (CNN) for target detection, which used saliency maps obtained with the modified Itti method as network input to improve the detection performance. In general, all these methods are designed with complex measurements and complex computational steps to reduce the negative effects of high sidelobes and background speckle noise of SAR images.

In recent years, sparse SAR imaging has achieved a superior performance in terms of the image quality over traditional matched filtering (MF)-based SAR imaging methods [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]. Sparse SAR imaging uses compressed sensing (CS) reconstruction algorithms to achieve high-quality recovery of sparse scenes with fewer data. Sparse reconstructed SAR images provide the advantages of low sidelobes and speckle noise, which has yielded new research directions in SAR applications [30]. However, current sparse SAR imaging methods must restore and compute an observation matrix with a large size, causing a high computational complexity. Thus, it is costly to obtain

Manuscript received 8 October 2022; revised 5 December 2022; accepted 9 January 2023. Date of publication 16 January 2023; date of current version 26 January 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62001508 and Grant 62131020. *(Corresponding author: Jia-Cheng Ni.)*

Jia-Cheng Ni, Ying Luo, Dan Wang, and Jia Liang are with the Institute of Information and Navigation, Air Force Engineering University, Xi'an 710077, China (e-mail: littlenjc@sina.com).

Qun Zhang is with the Institute of Information and Navigation and the Collaborative Innovation Center of Information Sensing and Understanding, Air Force Engineering University, Xi'an 710077, China, and also with the Key Laboratory for Information Science of Electromagnetic Waves, Ministry of Education, Fudan University, Shanghai 200433, China.

Digital Object Identifier 10.1109/TGRS.2023.3237632

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

sparse recovered SAR images directly from raw SAR echo data [25], [28].

In 2016, Bi et al. [31] proposed a novel regularization-based SAR image feature enhancement method using MF SAR images as input data, which attained a similar performance in reducing sidelobes and clutter to that of typical sparse SAR imaging methods. By using MF SAR images rather than raw echo data as input, the above sparse feature enhancement method does not require the calculation of an observation matrix, thus significantly reducing the computational complexity and allowing the algorithm to be processed in real time. SAR image feature enhancement can quickly reconstruct sparse images from a dataset of existing SAR images, which is highly valuable in sparse SAR target interpretation. Bi et al. [32] proposed a complex approximate message passing (CAMP) algorithm for sparse SAR feature enhancement, which obtained nonsparse estimates of considered scenes with complete image statistical characteristics. In other words, the CAMP algorithm could achieve CFAR detection of sparse SAR images. Recently, Bi et al. [33] used sparse SAR images as input data under a CNN architecture. This approach achieved a higher target classification performance than that obtained with the same network structure using MF SAR images as input. All the above studies focused on SAR interpretation using sparse feature enhancement SAR images. However, when faced with SAR target detection in complex backgrounds, current regularization-based SAR sparse feature enhancement methods suffer from the following drawbacks.

1) The sparsity and quality of the reconstruction image are directly influenced by the magnitude of the regularization parameter. If the parameter is not properly selected, the feature enhancement algorithm cannot obtain favorable target detection results. Since the observation scene is unknown, choosing the appropriate regularization parameters for SAR target detection remains a difficult problem [34], [35], [36].
2) It is difficult to evaluate the quality of reconstructed images. Existing methods mainly use image entropy and image contrast to evaluate the reconstruction quality. These algorithms often require a large number of iterations to obtain suitable evaluation indicators. However, in SAR target detection problems, even if the image entropy of the reconstructed image is low, this does not indicate that the algorithm can obtain a high target detection rate.

The above problems limit the application of SAR target detection using sparse SAR images. To date, there is no sparse feature enhancement method specifically designed for a particular target detection method. In addition, no research has focused on regularization parameter optimization, especially considering SAR target detection.

In this article, we attempted to increase the SAR target detection accuracy by using sparse feature enhancement images as input data. We designed a convolutional sparse feature enhancement method for SAR image saliency detection and established a modified superpixel Bayesian saliency target detector. We analyzed the specific requirements of Bayesian saliency detection for SAR images, which can be listed as follows: first, the speckle noise level of a given image should be low. Second, the intensity difference between the foreground and background of the image should be large. Third, the prior map must be as accurate as possible, and the superpixels corresponding to the target area must be consistent with the real target contour.

Following the above requirements, we proposed a convolutional sparse feature enhancement method specifically designed to meet the demands of Bayesian saliency maps. We added a convolution process to the objective function of L_1 regularization to reduce holes and preserve the contour of the target, all helpful in the calculation of superpixels. In addition, we proposed a regularization parameter optimization method to automatically obtain optimal regularization parameters for Bayesian saliency detection. The update process of the regularization parameter was correlated with the statistical characteristics of the input MF SAR image and changes in the Bayesian prior map. The proposed method could reduce the number of iterations while maintaining the target detection performance through parameter optimization.

The specific contributions of this article are as follows.

1) We proposed a saliency-based SAR target detection method using sparse feature enhancement-derived SAR images as input. We established a convolutional sparse feature enhancement method specifically designed for Bayesian saliency detection. Adding a convolution operation could reduce holes in the target area and better retain contour information of the target.
2) Through sparse feature enhancement, the image background could be effectively suppressed, and the pixel value of the image foreground could be averaged, resulting in a more accurate Bayesian prior saliency map. In addition, we could obtain better superpixel results by using only intensity information of sparse SAR images. There was no need to design a complex superpixel segmentation model based on the statistical characteristics of MF SAR images. Thus, the computational complexity of Bayesian saliency detection was reduced.
3) We proposed a regularization parameter optimization method for Bayesian saliency detection. The regularization parameters were updated iteratively, allowing the background noise to be quickly suppressed. The iteration process was stopped when the detection ability of the prior saliency map remained unchanged. This could greatly reduce the iteration number of the sparse feature enhancement process.

Through the above innovations, the proposed algorithm can effectively reduce the false alarm rate and improve the target detection performance, especially for images with sparse scenes. It should be noted that the proposed algorithm utilizes the sparse prior of the target, so the algorithm performance for very complex background images (the background contains many pixels with the same scattering coefficient as the target) may not be as good as that of sparse scenes.

This article is organized as follows. Section II introduces our convolutional sparse feature enhancement method.

Section III presents the Bayesian saliency detection method based on sparse feature enhancement images. Section IV describes the regularization parameter optimization method, algorithm steps, and parameter analysis approach. Section V provides the experimental and analysis results. Section VI concludes this article.

II. Modified Sparse Feature Enhancement Model

A. Sparse SAR Feature Enhancement Based on the Convolutional L_1 ADMM Algorithm

The classical model of SAR sparse feature enhancement can be expressed as [31]

$$\boldsymbol{Y}_{\mathrm{MF}} = \boldsymbol{X} + \boldsymbol{N} \tag{1}$$

where $\boldsymbol{Y}_{\mathrm{MF}} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ is the known MF SAR image, which exhibits a pixel size of $N_{\mathrm{a}} \times N_{\mathrm{r}}$, $\boldsymbol{X} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ is the feature enhancement result, and $\boldsymbol{N} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ is a matrix denoting the difference between the MF SAR image and feature enhancement-derived SAR image, including noise, clutter, and sidelobes. When $\boldsymbol{Y}_{\mathrm{MF}}$ is spatially sparse, $\boldsymbol{X}$ can be recovered by solving an L_p $(0 < p \leq 1)$ regularization optimization problem

$$\hat{\boldsymbol{X}} = \mathrm{argmin}_{\boldsymbol{X}} \left\{ \left\| \boldsymbol{Y}_{\mathrm{MF}} - \boldsymbol{X} \right\|_{\mathrm{F}}^2 + \lambda \left\| \boldsymbol{X} \right\|_p \right\} \tag{2}$$

where $\left\| \cdot \right\|_{\mathrm{F}}$ denotes the Frobenius norm of the matrix, $\left\| \cdot \right\|_p$ denotes the L_p-norm constraining the sparsity of the image, and we set $p = 1$ in the following derivation. λ is the regularization parameter. λ reflects the relationship between the similarity with $\boldsymbol{Y}_{\mathrm{MF}}$ and the sparsity of $\boldsymbol{X}$. Many iterative algorithms have been proposed to solve (2), including the iterative thresholding (IST) algorithm [37], approximate message passing (AMP) [38], and alternating direction of multipliers method (ADMM) [39].

The classic feature enhancement method only restricts the spatial sparsity of $\boldsymbol{X}$. As λ increases, the background noise in $\boldsymbol{X}$ significantly decreases. However, with increasing λ value, scattering points in the target area with a low scattering coefficient are also suppressed, resulting in many holes and discontinuities in the target area. This can destroy target contour information and cause difficulties in the subsequent target detection step.

To suppress the speckle noise to the greatest extent while maintaining the target contour, we included a convolution process in the second term of (2) and transformed the regularization problem into

$$\begin{aligned} \hat{\boldsymbol{X}} &= \mathrm{argmin}_{\boldsymbol{X}} \left\{ \left\| \boldsymbol{Y}_{\mathrm{MF}} - \boldsymbol{X} \right\|_{\mathrm{F}}^2 + \lambda \left\| \boldsymbol{W} \odot \boldsymbol{X} \right\|_1 \right\} \\ \boldsymbol{W} &= 1/(\mathrm{Conv}(\boldsymbol{X}, \boldsymbol{C}) + \varepsilon) \end{aligned} \tag{3}$$

where $\boldsymbol{W} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ denotes the convolution process, $\odot$ denotes the Hadamard product of matrices, and $\mathrm{Conv}(\boldsymbol{X}, \boldsymbol{C})$ denotes the matrix convolution operation between $\boldsymbol{X}$ and $\boldsymbol{C}$. $\boldsymbol{C}$ is a convolution kernel, and ε is a small constant. With a reasonable size and value of $\boldsymbol{C}$, pixels with a low scattering coefficient in the target area can be enhanced, while the background area near the target does not greatly change. In our convolution process, the sliding step was set to 1, and edge points were removed after convolution so that the size of $\boldsymbol{W}$ remained consistent with that of $\boldsymbol{X}$. Equation (3) is similar to the reweighted L_1 regularization method, which has been applied in inverse SAR (ISAR) sparse imaging [40], [41].

To solve (3), we used an L_1-based ADMM of the matrix multiplication form to constrain the optimization problem. Specifically, for $p = 1$, (3) can be rewritten as

$$\begin{aligned} &\min_{\boldsymbol{X},\boldsymbol{Z}} \left\{ \left\| \boldsymbol{Y}_{\mathrm{MF}} - \boldsymbol{X} \right\|_{\mathrm{F}}^2 + \lambda \left\| \boldsymbol{W} \odot \boldsymbol{Z} \right\|_1 \right\} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} \end{aligned} \tag{4}$$

where $\boldsymbol{Z} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ is an auxiliary matrix. The augmented Lagrange form of (4) can be derived as

$$\begin{aligned} L(\boldsymbol{X}, \boldsymbol{Z}, \boldsymbol{\beta}) = \frac{1}{2} \left\| \boldsymbol{Y}_{\mathrm{MF}} - \boldsymbol{X} \right\|_{\mathrm{F}}^2 + \lambda \left\| \boldsymbol{W} \odot \boldsymbol{Z} \right\|_1 + \langle \boldsymbol{\beta}, \boldsymbol{X} - \boldsymbol{Z} \rangle \\ + \frac{\rho}{2} \left\| \boldsymbol{X} - \boldsymbol{Z} \right\|_{\mathrm{F}}^2 \end{aligned} \tag{5}$$

where $\boldsymbol{\beta} \in \mathbb{C}^{N_{\mathrm{a}} \times N_{\mathrm{r}}}$ is the Lagrangian multiplier matrix, ρ is a penalty parameter, and $\langle \cdot, \cdot \rangle$ denotes the inner product of the matrix. Then, the ADMM can alternatively update $(\boldsymbol{X}, \boldsymbol{Z}, \boldsymbol{\beta})$ by solving the following three subproblems:

$$\begin{aligned} \boldsymbol{X}^{(k+1)} &= \arg\min_{\boldsymbol{X}} L\left(\boldsymbol{X}, \boldsymbol{Z}^{(k)}, \boldsymbol{\beta}^{(k)} \right) \\ &= \left(\boldsymbol{Y}_{\mathrm{MF}} - \boldsymbol{\beta}^{(k)} + \rho \cdot \boldsymbol{Z}^{(k)} \right) / (1 + \rho) \end{aligned} \tag{6}$$

$$\begin{aligned} \boldsymbol{Z}^{(k+1)} &= \arg\min_{\boldsymbol{Z}} L\left(\boldsymbol{X}^{(k+1)}, \boldsymbol{Z}, \boldsymbol{\beta}^{(k)} \right) \\ &= S\left(\boldsymbol{X}^{(k+1)} + \boldsymbol{\beta}^{(k)}; \lambda / \boldsymbol{T} \right) \end{aligned} \tag{7}$$

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} + \rho \left(\boldsymbol{X}^{(k+1)} - \boldsymbol{Z}^{(k+1)} \right) \tag{8}$$

where $k \in \{1, 2, \ldots, N_k\}$ denotes the kth iteration. In (6), $\boldsymbol{X}$ is iteratively updated by minimizing its gradient to zero. In (7), $\boldsymbol{Z}$ minimization entails a classic regularization problem and can be solved via the soft-thresholding method. $S(\cdot; \cdot)$ and $\boldsymbol{T}$ denote the soft-thresholding operator and threshold matrix, respectively, and expressed as follows:

$$\begin{aligned} S(x; y) &= \mathrm{sign}(x) \cdot \max(|x| - y, 0) \\ \boldsymbol{T} &= \rho \cdot \left| \mathrm{Conv}\left(\boldsymbol{X}^{(k+1)}, \boldsymbol{C} \right) + \varepsilon \right| \end{aligned} \tag{9}$$

where $\mathrm{sign}(\cdot)$ denotes the sigmoid function and $|\cdot|$ denotes the absolute value. The proposed convolutional L_1 ADMM algorithm could be established by alternatively updating (6)–(8) until convergence is reached.

B. Analysis of Parameters

There are some parameters that affect the performance of the algorithm and must be set appropriately. For initialization, we set $\boldsymbol{X}^{(0)} = \boldsymbol{Z}^{(0)} = \boldsymbol{\beta}^{(0)} = \boldsymbol{I}$, where $\boldsymbol{I}$ is the identity matrix. The total number of iterations was defined as $N_k = 100$. ε in (3) was set to $\varepsilon = 10^{-6}$.

Penalty parameter ρ controls the convergence speed of the iterative algorithm. This parameter is usually set to a number slightly smaller than 1 or can be iteratively adjusted via the residual balancing method [42]. For simplicity, we fixed $\rho = 0.99$.

Convolution kernel $\boldsymbol{C}$ controls the enhancement and continuity of the target area. The larger the size of $\boldsymbol{C}$, the fewer

holes in the target, but the smoother the target pixel value will be. The value of the center of $\boldsymbol{C}$ should not be less than the value of the edge area to prevent the pixel value of the target area from changing too much. In this article, our purpose was to fill holes inside the target area, so the size of $\boldsymbol{C}$ should be smaller than the target size. In this article, our purpose was to fill holes inside the target area, so the size of $\boldsymbol{C}$ should be smaller than the target size. Considering that the minimum radial dimension of vehicles and ships in a large scene SAR image is very small, we set $\boldsymbol{C}$ as follows:

$$\boldsymbol{C} = \begin{bmatrix} c_1 & c_1 & c_1 \\ c_1 & 1 & c_1 \\ c_1 & c_1 & c_1 \end{bmatrix} \tag{10}$$

where $c_1 \in [0.5, 1]$. We simply set $c_1 = 1$ in the rest of this article.

Regularization parameter λ is the most important parameter that largely affects the reconstruction results. In sparse SAR imaging under raw echo data, the cross-validation method is often still an acceptable choice. Zeng et al. [24] derived a solution assuming that the sparsity of the imaging scene is known. Zhang et al. [41] set λ related to the variance in the imaging scene. Fig. 1 shows two SAR images with complex clutter features and a comparison of different feature enhancement methods under different λ values. The second–fourth rows show the processing results obtained with the L_1 regularization method based on IST. When λ is small, the algorithm often failed to obtain satisfactory sparse enhancement results. When λ is large enough, the background area was suitably suppressed, but the ship target inevitably exhibited holes and discontinuities (refer to the fourth row). The contour of the ship target shown in Fig. 1(b4) was clearly discontinuous, and the ship target shown in Fig. 1(d4) contained many holes. It is difficult to select an appropriate λ value manually. The fifth–seventh rows of Fig. 1 show the feature enhancement results obtained with the proposed convolutional L_1 ADMM under different λ values. It could be observed that the hole points in the target area were filled to a great extent while the background was still suitably suppressed. We could also find that the value of λ should be much larger than that of the traditional L_1 regularization method to achieve a satisfactory performance. Nevertheless, it is difficult to determine which λ value obtains better images for target detection.

C. Analysis of Convergence

Another important problem is the convergence of algorithms. In SAR sparse feature enhancement, the common way to terminate the iteration process is to calculate the following objective function or allow the iteration number to reach $k \geq N_k$:

$$\left\| \boldsymbol{X}^{(k+1)} - \boldsymbol{X}^{(k)} \right\|_{\mathrm{F}} \leq \mathrm{Th} \tag{11}$$

where Th is a small constant, e.g., $\mathrm{Th} = 10^{-6}$. At least dozens of iterations are usually needed to obtain a satisfactory result. Fig. 2 shows the convergence results of the L_1 regularization and proposed methods to obtain Fig. 1(a4) and (a7), respectively. It could be observed that the L_1 regularization method only needed 12 iterations to reach convergence, while the proposed method needed 91 iterations to finally reach the termination condition. This occurred because the proposed method includes a convolution process in the second term of (4), causing $\boldsymbol{X}$ to change more notably in each iteration than that in the traditional method.

Throughout this section, the proposed method could suppress the background and enhance the target area (fill holes and maintain the target contour). However, the enhancement results obtained with the proposed method were still greatly affected by the selection of λ. This suggests that λ must be manually designed for different images, which is very inconvenient for the target detection task. In addition, the algorithm needed nearly 100 iterations to converge under the termination condition in (11), which is very time-consuming. Our purpose was to achieve fast and stable target detection, not a harsh convergence condition.

Next, we proposed a Bayesian saliency detection method based on the proposed feature enhancement method and established a regularization parameter optimization method to solve the existing problems.

III. Bayesian Saliency Detection Based on the Proposed Feature Enhancement Method

Through sparse feature enhancement, the image background was suitably suppressed, and the pixel value of the image foreground could be averaged, ensuring that Bayesian saliency detection represented a very suitable target detection method. In this section, we proposed a Bayesian saliency detection method based on sparse feature enhancement images.

Calculation of the Bayesian saliency map includes three steps.

Step 1: The input feature enhancement image was segmented into superpixels. This could preserve the shapes of the targets and reduce the calculation workload for large image scenes. The most commonly used simple linear iterative clustering (SLIC) method is not suitable for MF-based SAR images, and additional calculation procedures are usually included to enhance the suitability of the SLIC method for SAR images [43]. Tong et al. [44] introduced the generalized likelihood ratio dissimilarity measurement method to suppress speckle noise. Wang et al. [10] adopted location similarity in the SLIC method to maintain the target boundaries. In addition, Liu et al. [45] and Zhang et al. [46] proposed two adaptive superpixel generation algorithms which did not consider the SLIC method and speckle filtering. For our proposed feature enhancement SAR images, since speckle noise was greatly suppressed and the target contour was suitably maintained, we only used the Euclidean distance of the pixel intensity combined with the Euclidean distance of pixel positions to calculate SLIC superpixels. The similarity measurement D_{ij} between the cluster center j and given pixel i can be easily measured as follows:

$$D_{ij} = \sqrt{d_i^2 + d_s^2}$$

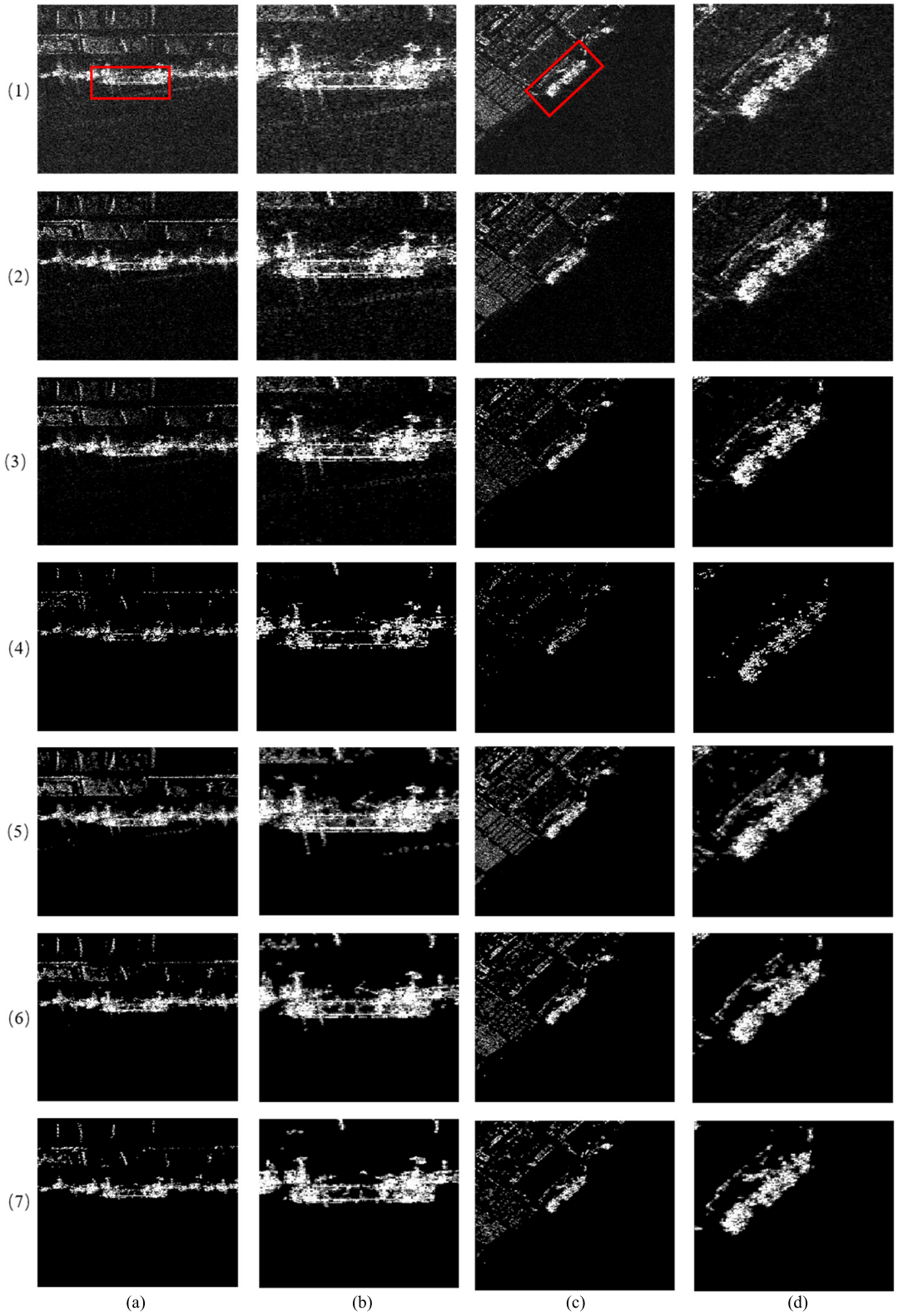


Fig. 1. (a1) MF SAR image of a ship target with a complex land background. (a2)–(a4) Feature enhancement results using the L_1 regularization method based on IST, with $\lambda = 25$, 80, and 230. (a5)–(a7) Feature enhancement results using the proposed convolutional L_1 ADMM, with $\lambda = 7 \times 10^3$, 1.4×10^4, and 5×10^4. (b) Zoomed-out view of the ship target in the red rectangle of image (a). (c1) MF SAR image of another ship target with a complex land background. (c2)–(c7) Feature enhancement results using the same algorithm and same λ values as those in (a). (d) Zoomed-out view of the ship target in (c).

$$d_i = \sqrt{l_i^2 + l_j^2}$$
$$d_s = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad (12)$$

where l_i and (x_i, y_i) denote the intensity and position, respectively, of pixel i. The other steps to calculate superpixels are the same as those in [43].

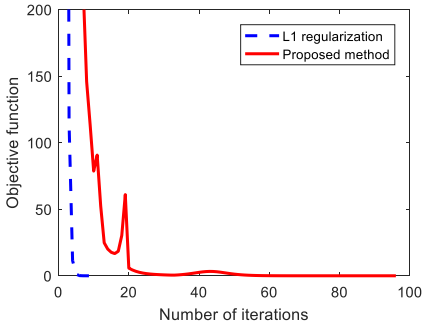


Fig. 2. Convergence performance of the L_1 regularization and proposed sparse feature enhancement methods.

Step 2: The Bayesian prior map was calculated using superpixels as the elementary units. The superpixels obtained in Step 1 are denoted as $s_i, i = 1, 2, \ldots, N_s$, where N_s is the total number of superpixels. The prior saliency map can be expressed as

$$P(\text{sal}|s_i) = \sum_{s_i, s_j \in N_s} \omega_{\text{pos}}(s_i, s_j) d_{\text{shape}}(s_i) d_{int}(s_i, s_j)$$
$$\omega_{\text{pos}}(s_i, s_j) = \exp\left(-\frac{d_{\text{pos}}(s_i, s_j)}{\sigma_{\text{pos}}^2}\right) \quad (13)$$

where $\omega_{\text{pos}}(s_i, s_j)$ denotes the distance weight and $d_{\text{pos}}(s_i, s_j)$ is the Euclidean distance between the centers of superpixels s_i and s_j. σ_{pos} is a position coefficient to control the contribution rate of the distance. We set $\sigma_{\text{pos}}^2 = 0.4$, the same as the setting in [43]. The pixel coordinates were normalized to [0, 1]. $d_{int}(s_i, s_j)$ denotes the Euclidean distance of the pixel intensity between s_i and s_j. $d_{\text{shape}}(s_i)$ denotes the number of pixels in s_i. After calculating (13), we obtained the normalized prior saliency map $\bar{P}(\text{sal}|s_i)$ via min-max scaling, as follows:

$$\bar{P}(\text{sal}|s_i) = \frac{P(\text{sal}|s_i) - P_{\min}(\text{sal}|s_i)}{P_{\max}(\text{sal}|s_i) - P_{\min}(\text{sal}|s_i)}. \quad (14)$$

After obtaining a normalized prior map of all superpixels, the total normalized prior map can be denoted as $\bar{P}(\text{sal}|s)$.

Step 3: Likelihood maps were generated. After obtaining the normalized Bayesian prior map, we divided the superpixels s_i into the potential image foreground fg and background bg as follows:

$$s_i \in \left\{ \begin{array}{ll} \text{fg}, & \text{when } \bar{P}(\text{sal}|s_i) \geq \text{mean}\left(\bar{P}(\text{sal}|s_i)\right) \\ \text{bg} & \text{otherwise} \end{array} \right\} \quad (15)$$

where mean() denotes the mean intensity of the prior map in s_i. Then, the observation likelihood can be computed as

$$P(z|\text{sal}) = \frac{N_{\text{fg}}(z)}{N_{\text{fg}}}$$
$$P(z|\text{bg}) = \frac{N_{\text{bg}}(z)}{N_{\text{bg}}} \quad (16)$$

where $P(z|\text{sal})$ and $P(z|\text{bg})$ denote the posterior probabilities that pixel z belongs to the saliency and background regions, respectively, $N_{\text{fg}}(z)$ and $N_{\text{bg}}(z)$ denote the number of pixels in the foreground area fg and background area bg, respectively, obtained with (12) with the same gray value as that of pixel z, and N_{sal} and N_{bg} denote the total pixel numbers of fg and bg, respectively.

Step 4: The Bayesian saliency map was obtained. Combining the prior map with the generated likelihood maps, the Bayesian saliency value of pixel z can be determined as

$$P(\text{sal}|z) = \frac{\bar{P}(\text{sal}|z) P(z|\text{sal})}{\bar{P}(\text{sal}|z) P(z|\text{sal}) + \left(1 - \bar{P}(\text{sal}|z)\right) P(z|\text{bg})} \quad (17)$$

where $\bar{P}(\text{sal}|z) = \bar{P}(\text{sal}|s_i)$ denotes pixel z, which belongs to superpixel s_i and has the same Bayesian saliency value.

In the Bayesian saliency map, the background was suitably suppressed, while the salient regions exhibited the average image intensity. Next, we could use a global threshold method to generate binary images for target detection, and the threshold could be easily selected.

IV. Regularization Parameter Optimization

A. Regularization Parameter Optimization

Here, we proposed a regularization parameter optimization method to achieve fast and accurate target detection. The core step to generate the Bayesian saliency map was to obtain a suitable prior map. The Bayesian prior map was related to the segmentation of superpixels and the intensity of the target pixels. It should be noted that in saliency-based target detection, final detection results are generally obtained by counting the large connected domains in the binary image. Our idea was to correlate λ with changes in the Bayesian prior saliency map, thus establishing a relationship between λ and the detection results. With increasing λ value, the target area in the feature enhancement images became increasingly prominent, causing changes in the Bayesian prior map. When the changes in a few large connected domains in the prior map were limited, the total Bayesian saliency map also remained basically unchanged. At this time, the target detection results together with the appropriate λ value could be obtained.

Specifically, we incorporated Steps 1–2 into our proposed convolutional L_1 ADMM algorithm and iteratively adjusted λ until the largest connected domains in the Bayesian prior map remained stable. We first set the lower bound of λ by analyzing the intensity histogram of the input MF SAR image. The left column of Fig. 3 shows three MF SAR images with different background types, and the right column shows their corresponding intensity histograms. When the number of high-intensity pixels in the image is small (such as the first two images), it could be generally considered that there occurred more background pixels and fewer target pixels in the image. At this time, it was necessary to set a relatively high initial value of λ to obtain a sparser feature enhancement image. Thus, the background was suppressed, and the target area was highlighted. When the number of high-intensity pixels in the image was large (such as the third SAR image), more targets did not necessarily occur in the image, but the foreground was more complex and contained more high-intensity pixels. At this time, to retain the targets in the foreground as much as possible, it was necessary to set a low initial value of λ.

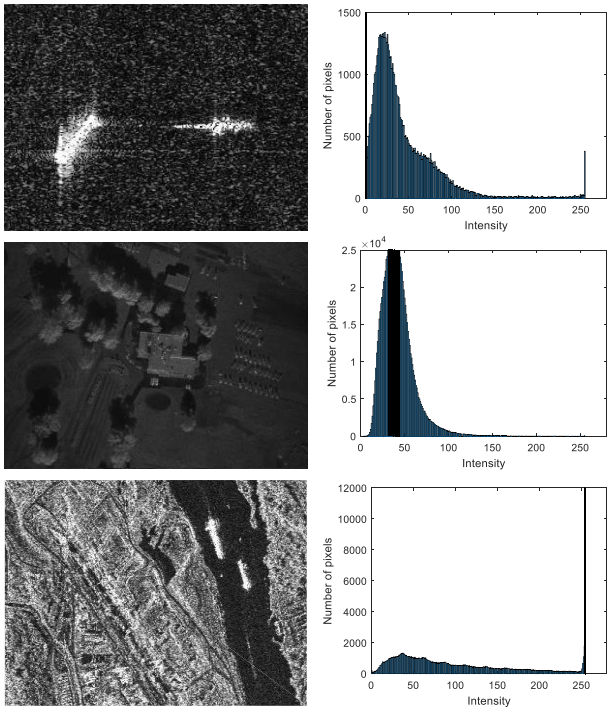


Fig. 3. MF SAR images with different background types and their corresponding intensity histograms. The image variance in the SAR images from top to bottom is 1.6×10^3, 336, and 5.9×10^3.

Here, we used the variance in the input MF SAR images as the base value of λ, denoted as $var(\boldsymbol{Y}_{\mathrm{MF}})$. The image variance reflects the intensity of the high-frequency part of the image, and it is positively correlated with the number of high-intensity pixels in the image.

We defined $\mathrm{sr} = N_{\mathrm{hi}}/N_{\mathrm{MF}}$ as the sparsity rate of the input images, where N_{hi} denotes the number of pixels with an intensity higher than 220 (the intensity value was quantified to range from 0 to 255). N_{MF} denotes the number of pixels in the whole MF SAR image. The lower the sr is, the fewer pixels in the foreground and the more pixels in the background, which suggests that a high initial value of λ must be set. Considering different sparsity rates, we iteratively increased λ relative to the initial value ini and the number of iterations k

$$\lambda^{(k)} = (\mathrm{ini} + k) \cdot \mathrm{var}(\boldsymbol{Y}_{\mathrm{MF}})$$
$$\mathrm{ini} = \left\{ \begin{array}{ll} 20 & \mathrm{sr} \leq 2\% \\ 10 & 2\% < \mathrm{sr} \leq 30\% \\ 5 & \mathrm{sr} > 30\% \end{array} \right\}. \tag{18}$$

B. Algorithm Steps

The proposed saliency detection algorithm is summarized in Algorithm 1. With increasing iteration k value, the proposed feature enhancement method could quickly suppress the background and highlight the target area, and the Bayesian saliency method could yield a satisfactory prior map. In Algorithm 1, the termination condition indicates that the largest connected domains in $\bar{P}(\mathrm{sal}|s)$ remained stable. We defined $\boldsymbol{P}_c^{(k)}$ as a binary map in the kth iteration, which only retained the top ten largest connected domains in $\bar{P}(\mathrm{sal}|s)$. The initial binary map could be obtained with (15), where the pixel value of fg was set to 1 and that of bg was set to 0. Our goal was to achieve fast and stable target detection rather than setting a harsh convergence condition to pursue a high reconstruction quality. Thus, we set a relatively high termination threshold as $\left\| \boldsymbol{P}_c^{(k+1)} - \boldsymbol{P}_c^{(k)} \right\|_{\mathrm{F}} \leq 10^{-3}$.

Algorithm 1 Bayesian Saliency Detection Based on Convolutional L_1 ADMM SAR Feature Enhancement

Input: $\boldsymbol{Y}_{\mathrm{MF}}$, C, N_k, ρ, ε,
Initialization: $\boldsymbol{X}^{(0)} = \boldsymbol{Z}^{(0)} = \boldsymbol{\beta}^{(0)} = \boldsymbol{P}_c^{(0)} = \boldsymbol{I}$, set ini by calculating sr based on the intensity histogram of $\boldsymbol{Y}_{\mathrm{MF}}$;
for $k = 1{:}N_k$
1) Update $\boldsymbol{X}^{(k+1)}$, $\boldsymbol{Z}^{(k+1)}$, $\boldsymbol{\beta}^{(k+1)}$ via (6)-(9)
2) Calculate superpixels:s_i, $i = 1, 2, \ldots, N_s$
3) Generate a normalized Bayesian prior map $\bar{P}(\mathrm{sal}|s)$ via (13)-(14)
4) Obtain binary image $\boldsymbol{P}_c^{(k+1)}$ only containing the top 10 largest connected domains in $\bar{P}(\mathrm{sal}|s)$.
if $\left\| \boldsymbol{P}_c^{(k+1)} - \boldsymbol{P}_c^{(k)} \right\|_F > 10^{-3}$ and $k < N_k$
let $k = k+1$, update λ via (18)
else break
Obtain a likelihood map through (15)-(16)
Generate a Bayesian map $P(\mathrm{sal}|z)$ via (17)
end
Output: Bayesian saliency map $P(\mathrm{sal}|z)$

Regarding the other parameters, we simply set them as constant values or assigned them the same values as those in the references. Their detailed values are provided in Section II.

C. Computational Complexity Analysis

The proposed saliency detection algorithm mainly comprises three steps, i.e., convolutional L_1 ADMM feature enhancement, superpixel calculation, and Bayesian saliency map generation. L_1 ADMM encompasses an elementwise operation, and its complexity is $\mathrm{O}(M \times I)$, where M is the number of pixels in the SAR image and I is the number of iterations. The computational complexity of superpixel calculation is $\mathrm{O}(M \times Q \times I)$, where Q is the number of pixels in the local patch centered at the pixel. We set $Q = 25$, the same setting as that in [16]. The computational complexity of Bayesian saliency map construction is $\mathrm{O}(N^2)$, where N is the number of superpixels. We set $N = M/300$. In saliency-based SAR target detection, the additional amount of computation was caused by L_1 ADMM feature enhancement, but reducing the number of iterations is very important.

V. Experimental Results

In this section, we demonstrated the effectiveness of the proposed method through a series of experiments. We first indicated the effectiveness of the proposed sparse feature enhancement method in saliency detection. Then, the target detection performance was evaluated through qualitative and quantitative comparisons to other target detection methods. Typical SAR images of sea and ground scenes with different sparsity and complex backgrounds were selected to evaluate

the performance of the proposed method. The sea scene images were retrieved from the SAR ship detection dataset (SSDD, which is an open dataset specifically designed for target detection and contains the ground truth. The ground scene images originated from the miniSAR real dataset acquired by the Sandia National Laboratories, Albuquerque, NM, USA, in which the ground truth was manually set. The reason we chose these two datasets is that both datasets contain multiple targets with complex backgrounds. Some background areas exhibited similar or even higher intensities than those of the targets. We considered the images in these two datasets sufficient to assess the performance of the proposed algorithm. All the experiments were conducted in MATLAB R2016 on a laptop with an Intel Core i5-5257 CPU (2.7 GHz) and 8 G of memory.

Fig. 4 shows the saliency detection results of a sea scene image using different methods. The input MF SAR image contained two ship targets and a wide land background area. The land background exhibited similar or even higher intensities than those of the ship targets, which could create notable difficulties in saliency detection. Fig. 4(a)–(d) shows that using the MF SAR image as input data could not yield satisfactory saliency detection results. Due to the complexity and high intensity of the background, the superpixel method failed to accurately extract the contour of the target, and the target area could not be highlighted in the Bayesian prior map. As shown in Fig. 4(e)–(l), we used sparse feature enhancement images obtained with the L_1-based IST method as input data. Two values of λ were chosen to assess the performance. Fig. 4(e)–(h) shows that although sea clutter was significantly suppressed by the L_1-based IST method, there remained many background superpixels with a high intensity. As a result, the target area in the prior map was not prominent enough, as was the target area in the final Bayesian prior map. Fig. 4(i)–(l) shows that if we selected a higher λ value, the feature enhancement result could become sparser. Since the L_1-based IST method could yield holes and discontinuities under the condition of high λ values, the superpixel method could not provide satisfactory segmentation results, especially in the ground background area (most superpixels in the background area were square). This led to poor results in generating the Bayesian prior map. Fig. 4(m)–(p) shows the results obtained with the proposed method. The initial value of λ was calculated as ini $= 5$, and five iterations were needed to reach convergence. The superpixel method could obtain a suitable target contour while segmenting the background region at the same time. The complete target area with a continuous ship contour could be obtained from the final Bayesian saliency map. Fig. 4(q)–(u) shows the changes in the top ten largest connected domains in $\boldsymbol{P}_c$. Only five iterations were needed in the proposed method to obtain the final saliency map, which saved time over the aforementioned methods needing nearly 100 iterations.

Fig. 5 shows the target detection result of another sea scene SAR image using the proposed method. The initial value of λ was calculated as ini $= 10$ through the corresponding intensity histogram, and six iterations were needed to achieve converge. Fig. 5(b) shows that although the algorithm chose a relatively high λ value, there occurred very few holes in the ship target area. As shown in Fig. 5(d), the foreground area was bright with a smooth and complete contour. In the Bayesian saliency map, the intensity difference between the foreground and background was large, and the pixel value of the foreground matched the average value. Fig. 5(e) shows the saliency map, and Fig. 5(f) shows the binary image obtained with a global threshold method. Fig. 5(h) shows the final detection results, where the minimum enclosing rectangles of the true targets are marked in green boxes. It could be observed that the proposed method could generate the precise contour of all ship targets with a lower false alarm rate.

Fig. 6 shows the target detection results of a miniSAR ground scene SAR image with a 0.1 m $\times$ 0.1 m resolution and 1638 $\times$ 2510 pixels. The image exhibited a complex foreground and background containing vehicles and other terrain features, such as buildings, roads, grasslands, and trees. Some terrain features exhibited similar or even higher intensities than those of the vehicles. The initial value of λ was calculated as ini $= 20$, which also indicates that the intensity difference between the background and foreground was small. Fig. 6(b) shows that roads and grasslands with a low intensity were suppressed, and the contour of the vehicles was retained. The proposed method needed seven iterations to obtain the final prior map. Fig. 6(d) and (e) shows that the foreground contained vehicle targets, buildings, and trees, where the contour of the target was smooth and complete. Unfortunately, it was difficult to extract only the vehicle targets from this SAR image without generating clutter-related false alarms. Fig. 6(f) shows the binary image after adding size information to remove clutter with a much larger size than that of the vehicles. Although false alarms still occurred in the binary image, all the target areas were suitably preserved. This is very helpful for the subsequent target feature extraction, especially to extract the target length, width, and other geometric features. Fig. 6(g) and (h) shows a binary image using comparison algorithms, in which we also included size information to remove large clutter. It could be found that the proposed method could retain the contour of the target to the highest extent. It should be noted that the binary map of the proposed method in Fig. 6 still remains some false alarms. It is because the proposed method applies the sparse prior information of the target in image. Therefore, when facing images with very complex background (the background contains many pixels with the same scattering coefficient as the target), the false alarm rate of the algorithm may not be as low as that of sparse scenes.

Fig. 7 shows the convergence performance of different feature enhancement methods when processing Figs. 5 and 6. Fig. 7(a) shows the convergence result of processing Fig. 5(a) using the L_1 regularization method, $L_{1/2}$ regularization method, convolutional L_1 ADMM, and convolutional L_1 ADMM after employing our regularization parameter optimization strategy. The proposed method needed six iterations to reach convergence, while the convolutional L_1 ADMM needed 100 iterations. This shows that the proposed regularization parameter optimization method could obtain similar target detection results with much fewer iterations. Fig. 7(b) shows the convergence result of processing Fig. 6(a),

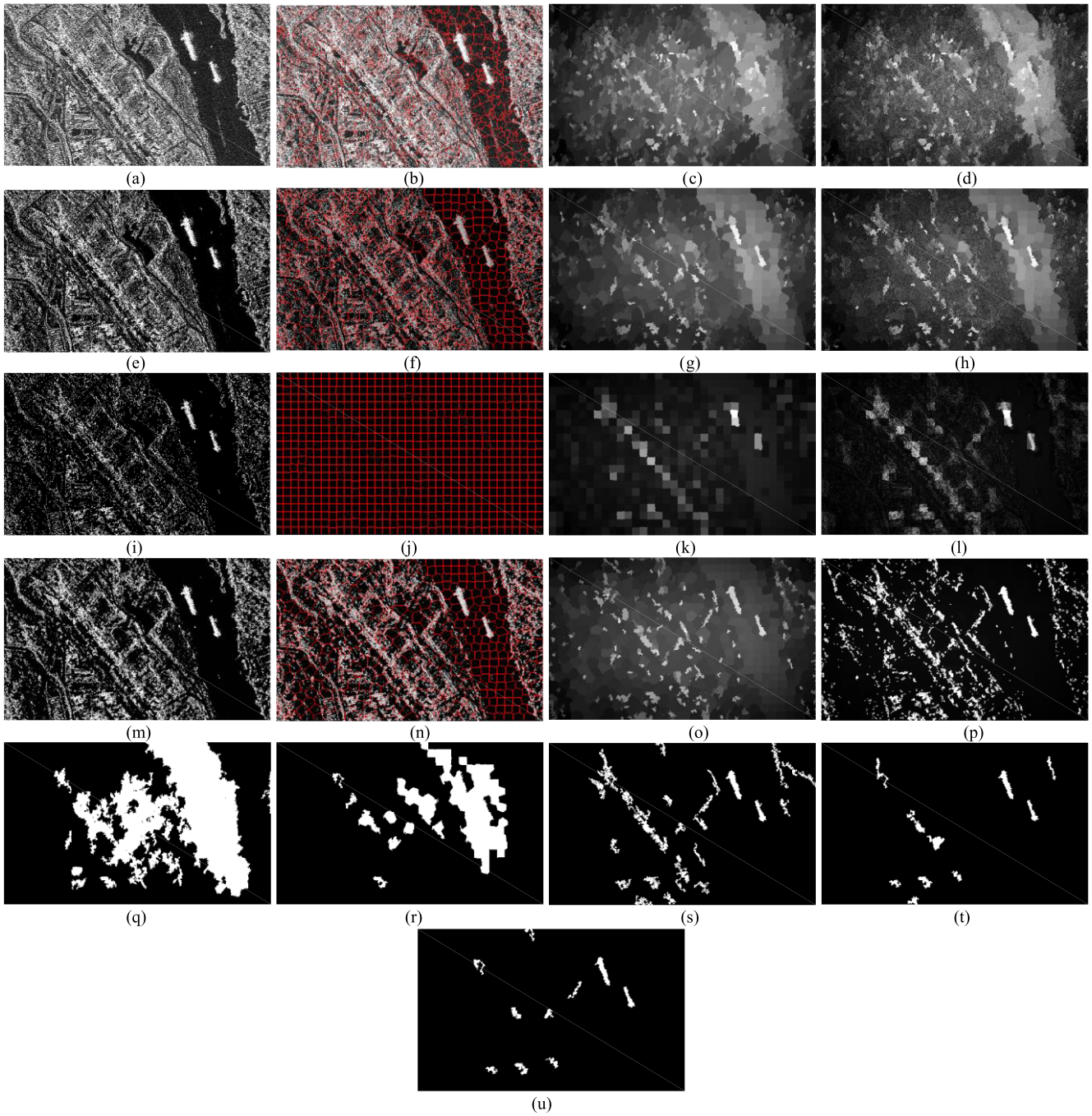


Fig. 4. Saliency detection results of different methods: (a)–(d) MF SAR images, corresponding superpixel results, Bayesian prior map, and Bayesian saliency map; (e)–(h) sparse feature enhancement image using L_1 regularization based on the IST method, with $\lambda = 50$, corresponding superpixel results, Bayesian prior map, and Bayesian saliency map; (i)–(l) sparse feature enhancement image using L_1 regularization based on the IST method, with $\lambda = 130$, corresponding superpixel results, Bayesian prior map, and Bayesian saliency map; (m)–(p) sparse feature enhancement image using the proposed convolutional L_1 ADMM, with ini $= 5$, corresponding superpixel results, Bayesian prior map, and Bayesian saliency map; and (q)–(u) changes in the top ten largest connected domains in $\boldsymbol{P}_c$.

which reveals the same experimental results. The proposed method only needed seven iterations to obtain the final prior map.

Next, we assessed the various detection methods, as shown in Figs. 5(a) and 6(a), to quantitatively evaluate the performance of our method. We compared the results obtained with the proposed method to the L_1-based IST recovered image and Bayesian saliency map (L_1+ BS), MF SAR image and Bayesian saliency map (MF + BS), MF SAR image and local CFAR [47], and MF SAR image and global CFAR [18]. Fig. 8 shows the corresponding receiver operating characteristic (ROC) curve. The closer to the upper left the ROC curve occurs, the higher the performance of the method. To compare methods, we set different parameters and attempted to obtain the best detection result. For (L_1+ BS), we set two λ values. In the CFAR-based method, as shown in Fig. 8(a), we set the

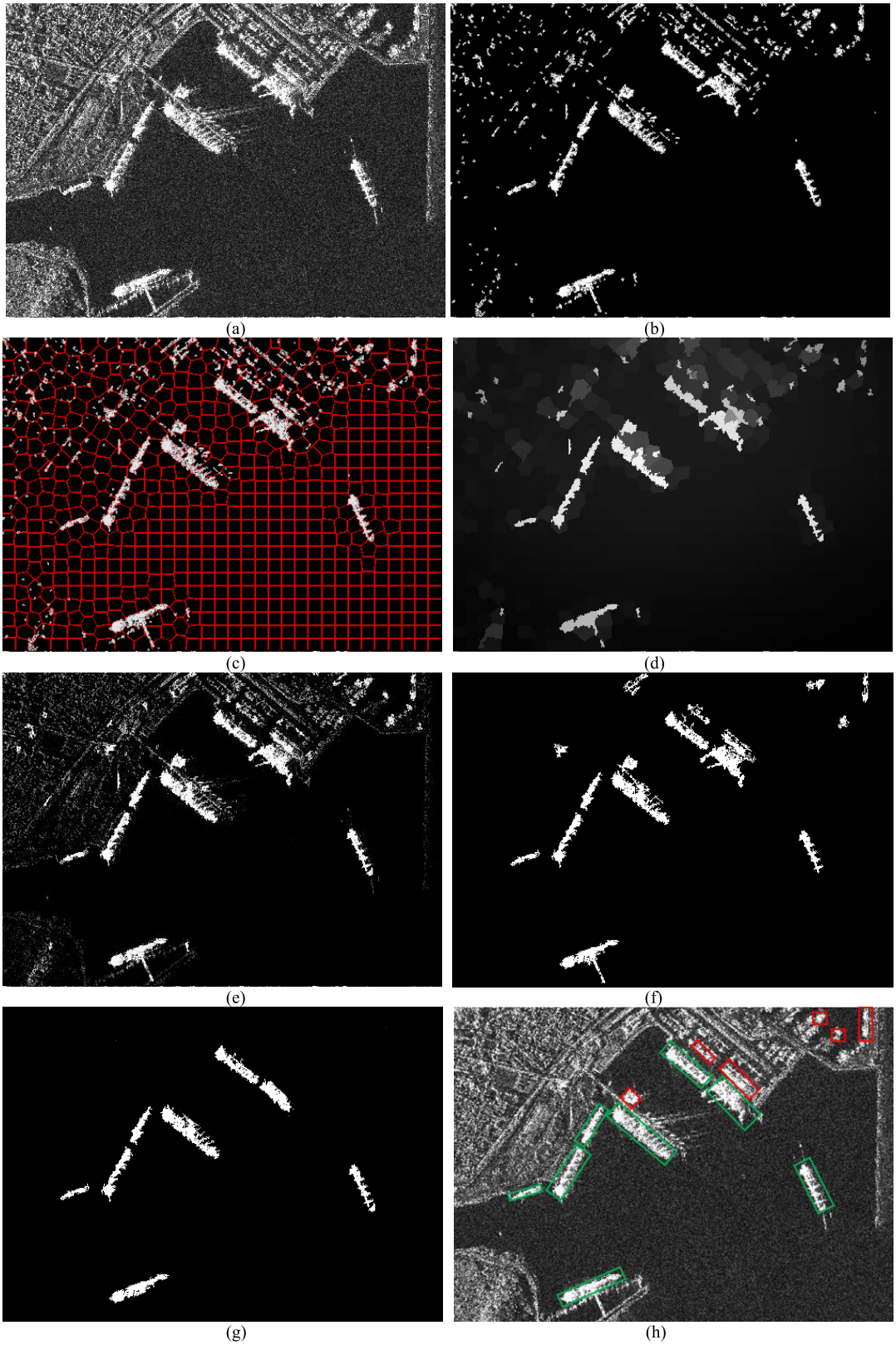


Fig. 5. Detection results using the proposed Bayesian saliency method: (a) input MF SAR images; (b) sparse feature enhancement image using the proposed convolutional L_1 ADMM, with ini $= 10$; (c) superpixel segmentation result; (d) Bayesian prior map after six iterations; (e) final Bayesian saliency map; (f) binary image using global threshold method; (g) ground truth; and (h) final detection result.

probability of a false alarm (PF) as PF $= 0.02$. As shown in Fig. 8(b), since PF $= 0.02$ could not yield enough target pixels, we set PF $= 0.1$. As shown in Fig. 8, the ROC curves obtained with the proposed method were better than those obtained with

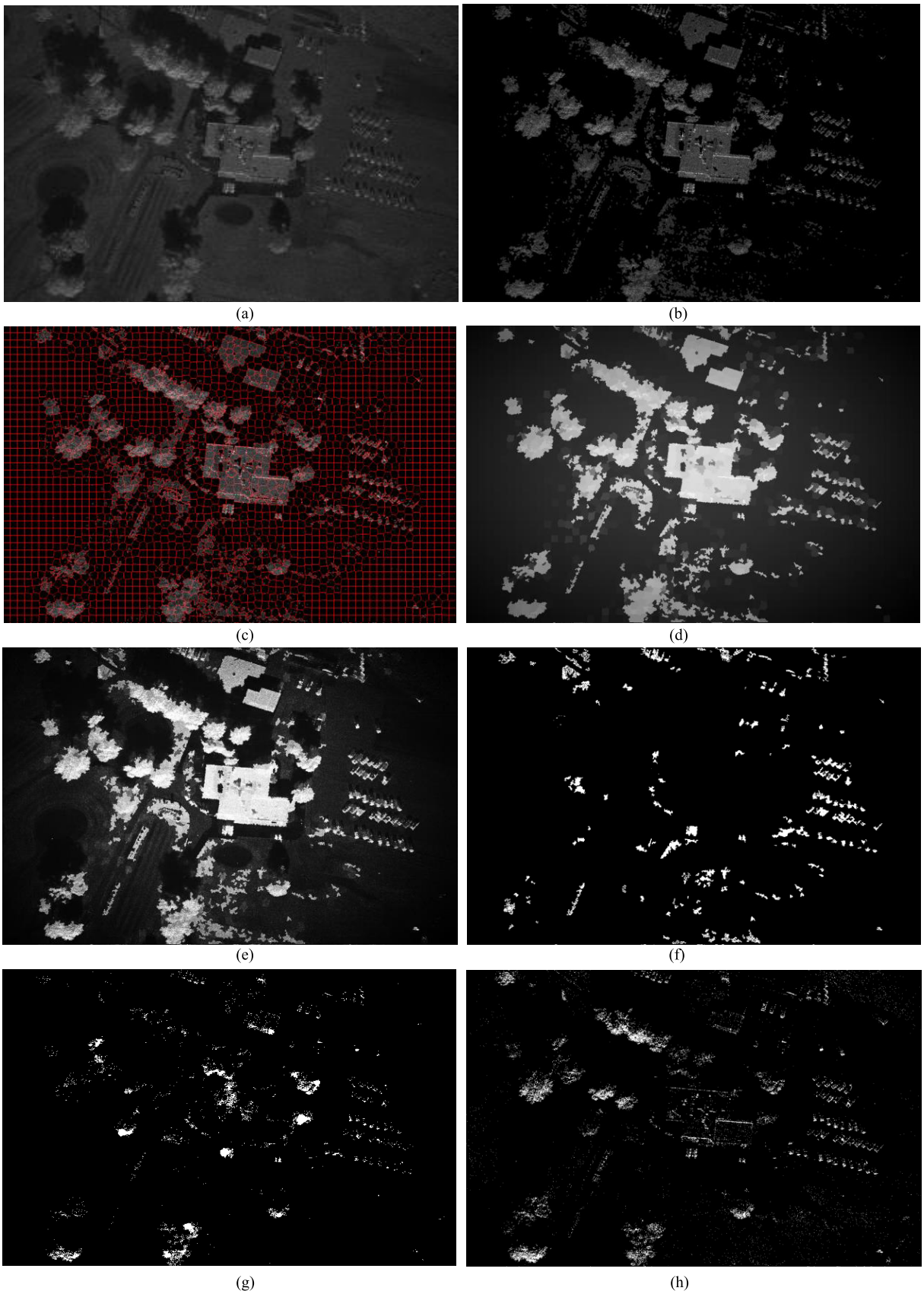


Fig. 6. Detection results using the proposed Bayesian saliency method: (a) input MF SAR images; (b) sparse feature enhancement image using the proposed convolutional L_1 ADMM, with ini $= 20$; (c) superpixel segmentation result; (d) Bayesian prior map after seven iterations; (e) final Bayesian saliency map; (f) binary image after adding size information; (g) binary image using the L_1-based IST method and Bayesian saliency map; and (h) binary image using MF + CFAR.

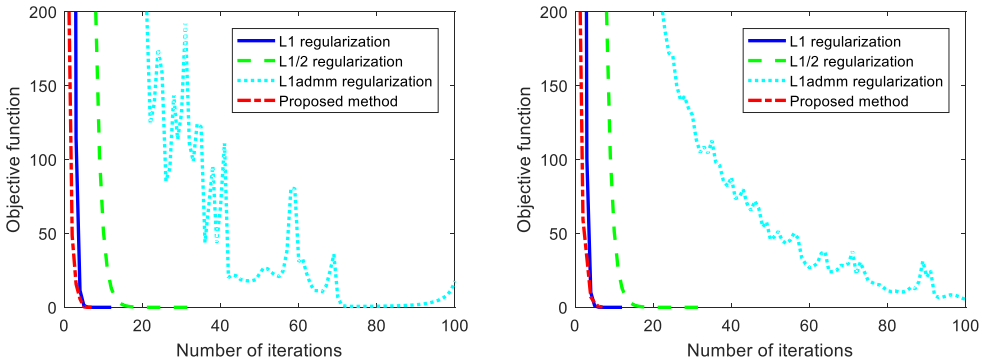


Fig. 7. (a) Convergence result of processing Fig. 5(a) using different algorithms. (b) Convergence result of processing Fig. 6(a) using different algorithms.

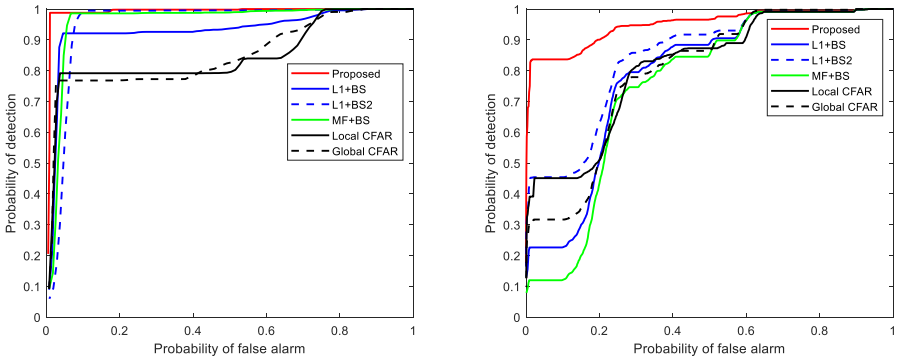


Fig. 8. ROC curves of different target detection methods: (a) image of Fig. 5(a): $\lambda = 50$ is set in L_1 + BS, $\lambda = 135$ is set in L_1 + BS2, and PF = 0.02 is set in local CFAR and global CFAR and (b) image of Fig. 6(a): $\lambda = 50$ is set in L_1 + BS, $\lambda = 75$ is set in L_1 + BS2, and PF = 0.1 is set in local CFAR and global CFAR.

the comparison methods, especially evident in Fig. 8(b), where the SAR image exhibited a complex background and a small intensity difference between the background and foreground.

As indicated in Table I, we further used the recovered SAR images obtained with the proposed convolutional L_1 ADMM as input and included two classic saliency methods, the Itti [16] and histogram contrast based salient region detection (HC) methods [17], to assess the detection performance. We denoted these two methods as CL_1+ Itti and CL_1 + HC, respectively. We selected mainstream criteria, such as the area under the curve (AUC), optimal operation point (T_{loss}), mean absolute error (MAE), precision, recall, and F-measure, to evaluate the performance of all methods. AUC describes the area under the ROC curve; the larger the AUC value is, the better the performance of the method. T_{loss} is the operation point that minimizes the following loss function using the probability of detection (PD) and PF:

$$T_{\text{loss}} = \text{PF} + \phi(1 - PD) \tag{19}$$

where $\phi = 1$ specifies the relative importance of false alarms (PF) and missed target loss (1 − PD). The smaller the T_{loss} is, the better the performance. MAE measures the average pixel difference between the binary map and the ground truth

$$\text{MAE} = \frac{1}{M}\sum_{i} |l_{i-\text{sm}} - l_{i-\text{gt}}| \tag{20}$$

where M is the number of pixels in the SAR image and $l_{i-\text{sm}}$ and $l_{i-\text{gt}}$ are the intensity value of the binary map and ground truth of pixel i, respectively. The smaller the MAE value is, the better the performance. We also employed the F-measure, which considers both the recall and precision

$$F_\gamma = \frac{(1+\gamma^2)\text{Precision} \times \text{Recall}}{\gamma^2 \text{Precision} + \text{Recall}} \tag{21}$$

where Precision measures the percentage of actual target pixels in the binary map, and Recall measures the percentage of actual target pixels detected. We set $\gamma^2 = 0.3$ to emphasize the precision, the same setting as that in [47].

Table I demonstrates that the proposed method outperformed the other methods, especially in image 2. The proposed method attained lower PF and T_{loss} values than those obtained with the other comparison methods, and the proposed method also attained a slightly higher PD value. CL_1 + Itti and CL_1 + HC also attained high PD values, but their PF value was higher than that of the proposed method. The proposed method attained the smallest MAE and largest AUC values in both images 1 and 2, further indicating that the detection performance of the proposed method was satisfactory for targets of different types in complex backgrounds. Table I reveals that the proposed method outperformed the other comparison methods in terms of the F-measure. The proposed

TABLE I
EXPERIMENTAL RESULTS FOR DIFFERENT CRITERIA

	METHOD	T_{loss}	MAE	AUC	Precision	Recall	F-measure
Image 1 (Fig. 5(a))	Proposed	**0.0305**	**0.0255**	**0.9902**	**0.6810**	**0.9013**	**0.7218**
	CL_1+Itti	0.0560	0.1345	0.9411	0.5510	0.8600	0.6008
	CL_1+HC	0.0660	0.1443	0.9135	0.5423	0.8517	0.5919
	L_1+BS	0.0610	0.0603	0.9300	0.4507	0.6931	0.4903
	L_1+BS2	0.0588	0.0544	0.9503	0.4216	0.7122	0.4654
	MF+BS	0.0820	0.0610	0.96	0.1933	0.7950	0.2342
	Local CFAR	0.0703	-	0.8472	0.1626	0.4220	0.1895
	Global CFAR	0.1030	-	0.8496	0.1388	0.5220	0.1671
Image 2 (Fig. 6(a))	Proposed	**0.0710**	**0.0433**	**0.9488**	**0.3890**	**0.8020**	**0.4415**
	CL_1+Itti	0.1220	0.1830	0.8735	0.3016	0.7652	0.3506
	CL_1+HC	0.1382	0.1993	0.8495	0.2905	0.7600	0.3388
	L_1+BS	0.1405	0.0910	0.7859	0.2710	0.5825	0.3092
	L_1+BS2	0.1336	0.0864	0.8438	0.2507	0.5530	0.2869
	MF+BS	0.2250	0.1020	0.7513	0.0964	0.7541	0.1207
	Local CFAR	0.2022	-	0.8119	0.0788	0.2520	0.0937
	Global CFAR	0.2366	-	0.7948	0.0726	0.1844	0.0844

method could detect most target pixels with fewer false alarms and false negative pixels. It reached the highest recall value, while the precision value was also high. The precision values of CL_1 + Itti and CL_1 + HC were lower than those of the proposed method, causing lower F-measure values. The recall values of L_1 + BS and CFAR were lower than that of the proposed method, also resulting in lower F-measure values. This occurs because in the L_1-based IST method with a high λ value, a large number of pixels in the vehicle target areas were removed with the background, which resulted in many holes in the target area. Therefore, many target pixels could not be detected. The proposed method could retain the target area via the convolutional L_1 ADMM and could reduce the background noise by setting a higher λ value. Thus, the proposed method outperformed the other methods. All the qualitative and quantitative experimental results verified that the proposed sparse feature enhancement method could improve the SAR target detection accuracy. The proposed Bayesian saliency-based SAR detection method obtained a higher detection performance than that of the traditional target detection methods.

VI. CONCLUSION

In this article, we aimed to improve the target detection performance using sparse feature enhancement SAR images as input and proposed a joint target detection method involving convolutional sparse feature enhancement and a Bayesian saliency detector. We first established a convolutional L_1 sparse regularization method to obtain sparse SAR images with continuous contours and fewer holes in the target area. Then, we developed a regularization parameter optimization method to quickly obtain optimal regularization parameters for saliency detection. The final detection result was obtained with a superpixel Bayesian saliency detector. Quantitative and qualitative experiments verified that the proposed method could provide the following advantages.

1) The proposed convolutional sparse SAR image feature enhancement method could better retain contour information of the target, thus facilitating superpixel calculation and improving the accuracy of the prior saliency map.
2) Through sparse feature enhancement, the image background was suitably suppressed, and the pixel value of the image foreground could be determined as the average value, which could reduce the computational complexity of the superpixel method. The proposed joint target detection method yielded better detection results than those obtained with traditional sparse regularization and target detection methods.
3) The proposed regularization parameter optimization method aimed to rapidly obtain an accurate Bayesian prior saliency map. This method could determine appropriate regularization parameter values while reducing the iteration number of the sparse feature enhancement method.

Further work will reduce the false alarms occurring in the method when the input SAR image exhibits a very complex background, and the computational efficiency of the algorithm will be improved for input images containing large scenes. Some new prior information like edge penalty can also be considered to add to algorithm to further improve the accuracy of superpixel segmentation.

ACKNOWLEDGMENT

The authors would like to thank the editors and anonymous reviewers for their valuable suggestions and comments that helped to greatly improve this article.

REFERENCES

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Apr. 2013.
[2] K. E. Darymli, P. McGuire, D. Power, and C. R. Moloney, "Target detection in synthetic aperture radar imagery: A state-of-the-art survey," *J. Appl. Remote Sens.*, vol. 7, no. 1, pp. 1–12, Jan. 2013.
[3] G. B. Goldstein, "False-alarm regulation in log-normal and Weibull clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-9, no. 1, pp. 84–92, Jan. 1973.
[4] D. Tao, S. N. Anfinsen, and C. Brekke, "Robust CFAR detector based on truncated statistics in multiple-target situations," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 117–134, Jan. 2016.

[5] W. Yu, Y. Wang, H. Liu, and J. He, "Superpixel-based CFAR target detection for high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 730–734, May 2016.

[6] S. Liu and Z. Cao, "SAR image target detection in complex environments based on improved visual attention algorithm," *EURASIP J. Wireless Commun. Netw.*, vol. 2014, no. 1, pp. 1–8, Dec. 2014.

[7] H. Wang, F. Xu, and S. Chen, "Saliency detector for SAR images based on pattern recurrence," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 2891–2900, Jul. 2016.

[8] W. Ni, L. Ma, W. Yan, H. Zhang, and J. Wu, "Background context-aware-based SAR image saliency detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1392–1396, Sep. 2018.

[9] S. Tu and Y. Su, "Fast and accurate target detection based on multiscale saliency and active contour model for high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5729–5744, Oct. 2016.

[10] Z. Wang, L. Du, and H. Su, "Target detection via Bayesian-morphological saliency in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5455–5466, Oct. 2017.

[11] L. Du, L. Li, D. Wei, and J. Mao, "Saliency-guided single shot multibox detector for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3366–3376, May 2020.

[12] M. Kang, Z. Lin, X. Leng, and K. Ji, "A modified faster R-CNN based on CFAR algorithm for SAR ship detection," in *Proc. RSIP Workshop*, May 2017, pp. 1–4.

[13] N. Wang, Y. Wang, H. Liu, Q. Zuo, and J. He, "Feature-fused SAR target discrimination using multiple convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1695–1699, Oct. 2017.

[14] J. Tang, J. Cheng, D. Xiang, and C. Hu, "Large-difference-scale target detection using a revised Bhattacharyya distance in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] S. Zhao, Z. Zhang, W. Guo, and Y. Luo, "An automatic ship detection method adapting to different satellites SAR images with feature alignment and compensation loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[16] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998.

[17] M.-M. Cheng, N. J. Mitra, X. Huang, P. H. S. Torr, and S.-M. Hu, "Global contrast based salient region detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 569–582, Mar. 2015.

[18] X. Li, H. Lu, L. Zhang, X. Ruan, and M.-H. Yang, "Saliency detection via dense and sparse reconstruction," in *Proc. 14th IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2976–2983.

[19] H. J. Seo and P. Milanfar, "Static and space-time visual saliency detection by self-resemblance," *J. Vis.*, vol. 9, no. 12, pp. 1–27, 2009.

[20] M. T. Alonso, P. Lopez-Dekker, and J. Mallorqui, "A novel strategy for radar imaging based on compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 12, pp. 4285–4295, Dec. 2010.

[21] L. C. Potter, E. Ertin, J. T. Parker, and M. Cetin, "Sparsity and compressed sensing in radar imaging," *Proc. IEEE*, vol. 98, no. 6, pp. 1006–1020, Jun. 2010.

[22] N. Ö. Onhon and M. Cetin, "A sparsity-driven approach for joint SAR imaging and phase error correction," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2075–2088, Apr. 2012.

[23] B. Zhang, W. Hong, and Y. Wu, "Sparse microwave imaging: Principles and applications," *Sci. China Inf. Sci.*, vol. 55, no. 8, pp. 1722–1754, 2012.

[24] S. Zeng, J. Fang, and B. Xu, "Sparse SAR imaging based on $L_{1/2}$ regularization," *Sci China Inf. Sci.*, vol. 55, no. 8, pp. 1755–1775, Aug. 2012.

[25] J. Fang, Z. Xu, B. Zhang, W. Hong, and Y. Wu, "Fast compressed sensing SAR imaging based on approximated observation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 352–363, Jan. 2014.

[26] J.-C. Ni, Q. Zhang, Y.-F. Yin, Y. Luo, and L. Sun, "A novel scan SAR imaging method for maritime surveillance via L_p regularization," *Int. J. Remote Sens.*, vol. 39, no. 1, pp. 169–190, Jan. 2018.

[27] J. Ni et al., "Compressed sensing SAR imaging based on centralized sparse representation," *IEEE Sensors J.*, vol. 18, no. 12, pp. 4920–4932, Apr. 2018.

[28] H. Bi, G. Bi, B. Zhang, W. Hong, and Y. Wu, "From theory to application: Real-time sparse SAR imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2928–2936, Apr. 2020.

[29] Z. Xu et al., "Sparse SAR imaging and quantitative evaluation based on nonconvex and TV regularization," *Remote Sens.*, vol. 13, no. 9, pp. 1–12, Nov. 2021.

[30] Z. Zhang and W. X. Yu, "Feature understanding and target detection for sparse microwave synthetic aperture radar images," *J. Radars.*, vol. 5, no. 1, pp. 42–56, Jun. 2016.

[31] H. Bi, B. Zhang, Z. Wang, and W. Hong, "L_q regularisation-based synthetic aperture radar image feature enhancement via iterative thresholding algorithm," *Electron. Lett.*, vol. 52, no. 15, pp. 1336–1338, Jul. 2016.

[32] H. Bi, B. C. Zhang, and X. X. Zhu, "L_1-regularization-based SAR imaging and CFAR detection via complex approximated message passing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3426–3440, 2017.

[33] H. Bi et al., "CNN-based target detection and classification when sparse SAR image dataset is available," *IEEE J. Sel. Topics. Appl. Earth Observ. Remote Sens.*, vol. 14, no. 1, pp. 6815–6827, Jan. 2021.

[34] Z. Xu, X. Chang, F. Xu, and H. Zhang, "$L_{1/2}$ regularization: A thresholding representation theory and a fast solver," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 7, pp. 1013–1028. Jul. 2012.

[35] S. Peng, C. Qu, J. Li, and Z. Li, "Regularization parameter estimation for point-based synthetic aperture radar image feature enhancement based on Mellin transform," *J. Appl. Remote Sens.*, vol. 11, no. 4, pp. 1–13. Apr. 2017.

[36] H. Shen, L. Peng, L. Yue, Q. Yuan, and L. Zhang, "Adaptive norm selection for regularized image restoration and super-resolution," *IEEE Trans. Cybern.*, vol. 46, no. 6, pp. 1388–1399, Jun. 2016.

[37] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. 27th Int. Conf. Int. Conf. Mach. Learn.*, Jun. 2010, pp. 399–406.

[38] D. L. Donoho, A. Maleki, and A. Montanari, "Message-passing algorithms for compressed sensing," *Proc. Nat. Acad. Sci. USA*, vol. 106, no. 45, pp. 18914–18919, Jul. 2009.

[39] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, Jan. 2011.

[40] E. Candes, M. Wakin, and S. Boyd, "Enhancing sparsity by reweighted 11 minimization," *J. Fourier Anal. Appl.*, vol. 14, no. 5, pp. 877–905, May 2008.

[41] S. Zhang, Y. Liu, X. Li, and D. Hu, "Enhancing ISAR image efficiently via convolutional reweighted L_1 minimization," *IEEE Trans. Image. Procss.*, vol. 30, no. 4, pp. 4291–4305, Apr. 2021.

[42] B. S. He, H. Yang, and S. L. Wang, "Alternating direction method with self-adaptive penalty parameters for monotone variational inequalities," *J. Optim. Theory Appl.*, vol. 106, no. 2, pp. 337–356, 2000.

[43] R. Achanta et al., "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, May 2012.

[44] N. Tong, H. Lu, L. Zhang, and X. Ruan, "Saliency detection with multi-scale superpixels," *IEEE Signal Process. Lett.*, vol. 21, no. 9, pp. 1035–1039, Sep. 2014.

[45] J. Liu, Q. Wang, J. Cheng, D. Xiang, and W. Jing, "Multitask learning-based for SAR image superpixel generation," *Remote Sens.*, vol. 14, 899, pp. 1–17, Feb. 2022.

[46] L. Zhang, S. Lu, C. Hu, D. Xiang, T. Liu, and Y. Su, "Superpixel generation for SAR imagery based on fast DBSCAN clustering with edge penalty," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 804–819, 2022.

[47] G. Gao, G. Kuang, Q. Zhang, and D. Li, "Fast detecting and locating groups of targets in high-resolution SAR images," *Pattern Recognit.*, vol. 40, no. 4, pp. 1378–1384, Apr. 2007.



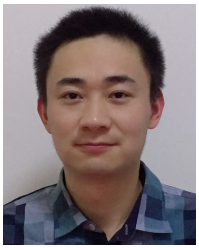
Jia-Cheng Ni was born in Shaanxi, China, in 1990. He received the M.S. and Ph.D. degrees in electrical engineering from Air Force Engineering University (AFEU), Xi'an, China, in 2014 and 2018, respectively.

He is currently with the Radar and Signal Processing Laboratory, Institute of Information and Navigation, AFEU, and also with the Collaborative Innovation Center of Information Sensing and Understanding. His research interests include synthetic aperture radar (SAR) sparse signal processing, SAR imaging, and auto target recognition (ATR).



Ying Luo (Member, IEEE) was born in Hunan, China, in 1984. He received the M.S. degree in electrical engineering from the Institute of Telecommunication Engineering, Air Force Engineering University (AFEU), Xi'an, China, in 2008, and the Ph.D. degree in electrical engineering from AFEU, in 2013.

He was a Post-Doctoral Fellow with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, from 2014 to 2017, and a Visiting Scholar with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore, from 2017 to 2018. He is currently a Professor and a Ph.D. Advisor with the Institute of Information and Navigation, AFEU. He has authored three books and over 100 articles on journals and conferences. Two of these articles won the First-grade Prize of Shaanxi Natural Science Excellent Academic Paper in 2010 and 2013, respectively. His research interests include signal processing and auto target recognition (ATR) in synthetic aperture radar (SAR) and inverse SAR (ISAR).



Dan Wang was born in Anhui, China, in 1990. He received the M.S. degree in information and communication engineering from the College of Electronic Science and Engineering, National University of Defense Technology, Changsha, China, in 2013. He is currently pursuing the Ph.D. degree in electrical engineering with the Institute of Information and Navigation, Air Force Engineering University (AFEU), Xi'an, China.

He is currently with the Radar and Signal Processing Laboratory, Institute of Information and Navigation, AFEU, and also with the Collaborative Innovation Center of Information Sensing and Understanding. His research interests include signal processing and radar imaging.



Jia Liang was born in Xi'an, China, in 1985. He received the M.A. degree in information and communication engineering from Air Force Engineering University, Xi'an, in 2016, where he is currently pursuing the Ph.D. degree in signal and information processing.

He has authored over 20 articles on journals and conferences. His research interests include array radar signal processing and target feature extraction and recognition.



Qun Zhang (Senior Member, IEEE) received the M.S. degree in mathematics from Shaanxi Normal University, Xi'an, China, in 1988, and the Ph.D. degree in electrical engineering from Xidian University, Xi'an, in 2001.

He was a Research Engineer from 2001 to 2003 and a Research Fellow from 2005 to 2006 with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is currently a Professor with the Institute of Information and Navigation, Air Force Engineering University, Xi'an, and an Adjunct Professor with the Key Laboratory for Information Science of Electromagnetic Waves, Ministry of Education, Fudan University, Shanghai, China. He has authored over 200 articles on journals and conferences. His main research interests include signal processing, clutter suppression, and its application in synthetic aperture radar (SAR) and inverse SAR (ISAR).